

Tribhuvan University
Faculty of Humanities and Social Science
Prithvi Narayan Campus, Pokhara

Lab Report on Computer Networking (CACS303)



BCA 5th Sem

[2079 Batch]

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Batch: 2079

Submitted To

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BCA Program, PNC

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S.N.	Objective	Date of submission	Remarks
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If you are already registered please enter your login information below:

Username :	test
Password :	
login	

You can also [signup here](#).

Signup disabled. Please use the username **test** and the password **test**.

No.	Time	Source	Destination	Protocol	Length	Destination Address	Info
20	5.610561	192.168.1.86	44.228.249.3	HTTP	699	44.228.249.3	POST /userinfo.php HTTP/1.1 (application/x-www-form-urlencoded)
21	5.938091	44.228.249.3	192.168.1.86	TCP	1466	192.168.1.86	80 → 50599 [ACK] Seq=1 Ack=646 Win=472 Len=1412 [TCP PDU reassembled in 23]
22	5.938091	44.228.249.3	192.168.1.86	TCP	1466	192.168.1.86	80 → 50599 [PSH, ACK] Seq=1413 Ack=646 Win=472 Len=1412 [TCP PDU reassembled in 23]
23	5.938091	44.228.249.3	192.168.1.86	HTTP	157	192.168.1.86	HTTP/1.1 200 OK (text/html)
24	5.938201	192.168.1.86	44.228.249.3	TCP	54	44.228.249.3	50599 → 80 [ACK] Seq=646 Ack=2928 Win=259 Len=0
85	8.494698	192.168.1.86	44.228.249.3	TCP	55	44.228.249.3	50598 → 80 [ACK] Seq=1 Ack=1 Win=512 Len=1
105	9.575765	44.228.249.3	192.168.1.86	TCP	66	192.168.1.86	80 → 50598 [ACK] Seq=1 Ack=2 Win=485 Len=0 SLE=1 SRE=2
135	10.757214	192.168.1.86	44.228.249.3	TCP	54	44.228.249.3	50598 → 80 [FIN, ACK] Seq=2 Ack=1 Win=512 Len=0
136	10.757413	192.168.1.86	44.228.249.3	TCP	54	44.228.249.3	50599 → 80 [FIN, ACK] Seq=646 Ack=2928 Win=259 Len=0

```
Internet Protocol Version 4, Src: 192.168.1.86, Dst: 44.228.249.3
Transmission Control Protocol, Src Port: 50599, Dst Port: 80, Seq: 646, Win: 259, Len: 0
Hypertext Transfer Protocol
HTML Form URL Encoded: application/x-www-form-urlencoded
  Form item: "uname" = "test"
  Form item: "pass" = "test"
```

Conclusion

This lab demonstrates the security risk of using unencrypted protocols for transmitting sensitive information. Attackers on the same network can easily intercept login credentials if encryption is not used.

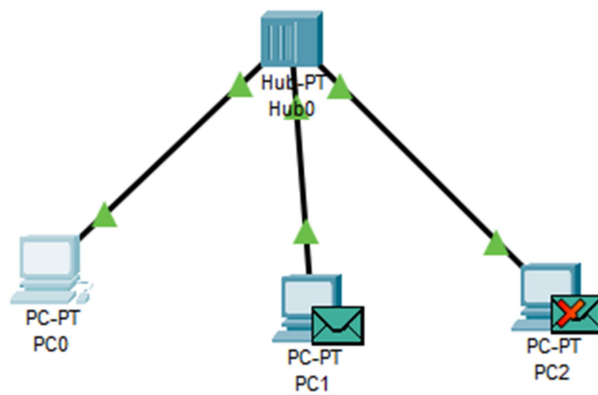
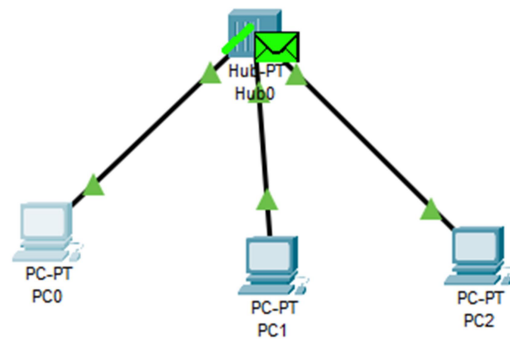
Command Prompt

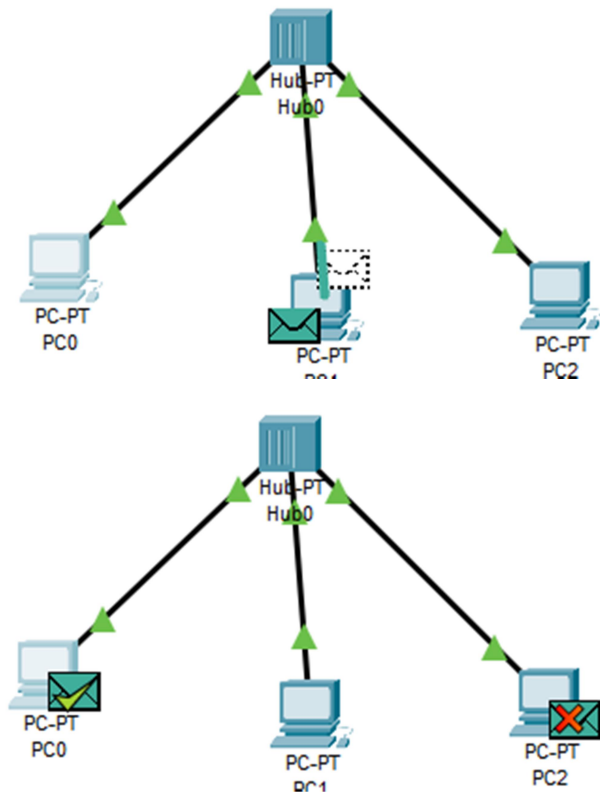
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

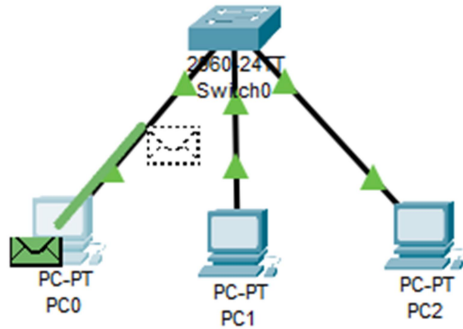




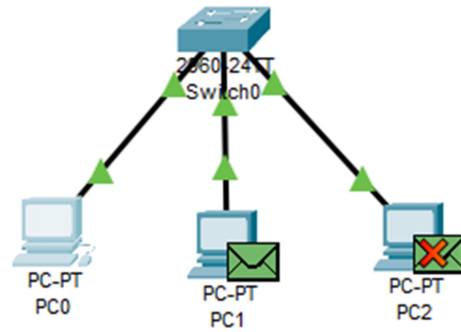
Conclusion

The Hub, a Layer 1 device, operates by broadcasting signals to all ports, leading to increased traffic and possible collisions. This behavior was clearly demonstrated using Cisco Packet Tracer, helping visualize the limitations and working mechanism of hubs in real-time.

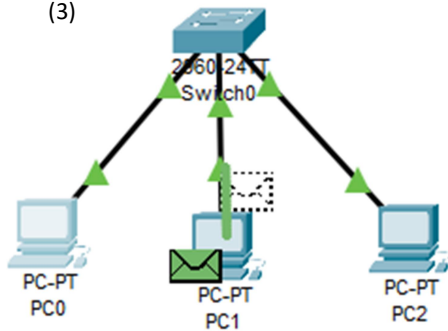
(1)



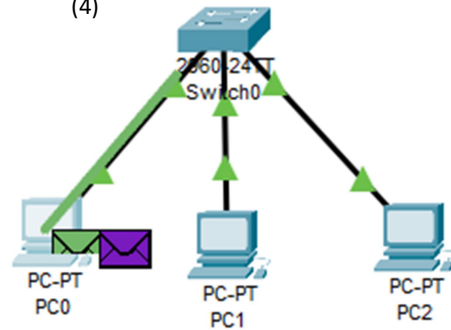
(2)



(3)



(4)



Conclusion

This lab demonstrated that a switch intelligently forwards traffic to the correct destination using MAC address tables. Unlike a hub, it reduces collisions and increases efficiency, making it ideal for modern LANs.